# MAT 549 HW03

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**Problem 1** (10-6, vector bundle construction theorem)**.** Let $M$ be a smooth manifold and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of $M$. Suppose that for each $\alpha, \beta \in A$ we are given a smooth map $\tau_{\alpha\beta} \colon U_\alpha \cap U_\beta \to GL_k\mathbb{R}$ such that $\tau_{\alpha\beta}\tau_{\beta\gamma} = \tau_{\alpha\gamma}$ on $U_\alpha \cap U_\beta \cap U_\gamma$ for all $\alpha, \beta, \gamma \in A$. Show that there is a smooth rank-$k$ vector bundle $p \colon E \to M$ with smooth local trivializations $\Phi_\alpha \colon p^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose transition functions are the given maps $\tau_{\alpha\beta}$.

*Solution.* Let $E = (\coprod_{\alpha \in A} U_\alpha \times \mathbb{R}^k)/\sim$ where $(x_1, v_1) \in U_\alpha \times \mathbb{R}_k$ and $(x_2, v_2) \in U_\beta \times \mathbb{R}^k$ satisfy $(x_1, v_1) \sim (x_2, v_2)$ if $(x_1, v_1) = (x_2, \tau_{\alpha\beta}(x_1)v_2)$ (in particular $x_1$ must equal $x_2$). Observe that $\sim$ is an equivalence relation:

- Let $(x, v) \in U_\alpha \times \mathbb{R}^k$. Note that $\tau_{\alpha\alpha}(x) = \mathrm{id}_{x_\alpha \times \mathbb{R}^k}$ since $\tau_{\alpha\alpha}(x)^2 = \tau_{\alpha\alpha}(x)$ for each $x \in U_\alpha$ by the cocycle condition and the invertibility of $\tau_{\alpha\alpha}(x) \in GL_k\mathbb{R}$, so $(x, v) = (x, \tau_{\alpha\alpha}(x)v)$ Thus $(x, v) \sim (x, v)$.
- Let $(x, v_1), (x, v_2) \in (U_\alpha \cap U_\beta) \times \mathbb{R}^k$ such that $(x, v_1) \sim (x, v_2)$. Then observe that
$$\tau_{\alpha\beta}(x)\tau_{\beta\alpha}(x) = \tau_{\alpha\alpha}(x) = \mathrm{id}_{x \times \mathbb{R}^k}, \qquad \tau_{\beta\alpha}(x)\tau_{\alpha\beta}(x) = \tau_{\beta\beta}(x) = \mathrm{id}_{x \times \mathbb{R}^k}$$
for $x \in U_\alpha \cap U_\beta$, so $\tau_{\alpha\beta}(x)$ and $\tau_{\beta\alpha}(x)$ are inverse linear transformations. Thus
$$v_1 = \tau_{\alpha\beta}(x)v_2 \iff v_2 = \tau_{\beta\alpha}(x)v_1$$
which means that $(x, v_2) \sim (x, v_1)$.
- Let $(x, v_1), (x, v_2), (x, v_3) \in (U_\alpha \cap U_\beta \cap U_\gamma) \times \mathbb{R}^k$, such that $(x, v_1) \sim (x, v_2)$ and $(x, v_2) \sim (x, v_3)$. Then
$$v_1 = \tau_{\alpha\beta}(x)v_2 = \tau_{\alpha\beta}(x)\tau_{\beta\gamma}(x)v_3 = \tau_{\alpha\gamma}(x)v_3$$
by the cocycle condition, so $(x, v_1) \sim (x, v_3)$.

Define $p \colon E \to M$ by sending the equivalence class of a pair $(x, v)$ to $x$; this is well-defined because two equivalent elements in $\coprod_{\alpha \in A} U_\alpha \times \mathbb{R}^k$ must have the same first coordinate by definition. Then, define
$$\Phi_\alpha \colon p^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k, \qquad \Phi_\alpha([(x, v)]) = (x, v)$$
where $[(x, v)]$ is the equivalence class of $(x, v) \in U_\alpha \times \mathbb{R}^k$; this is well-defined because if $(x, v_1) \sim (x, v_2)$ for $(x, v_1), (x, v_2) \in U_\alpha \times \mathbb{R}^k$, then $v_1 = \tau_{\alpha\alpha}(x)v_2 = v_2$, i.e. each equivalence class in $E$ contains at most one element of each set in the disjoint union, and it is a bijection because it has an inverse given by just sending $(x, v)$ to the equivalence class containing it. We can then see that for $(x, v) \in (U_\alpha \cap U_\beta) \times \mathbb{R}^k$,
$$\Phi_\alpha \circ \Phi_\beta^{-1}(x, v) = \Phi_\alpha([(x, v)]) = \Phi_\alpha([(x, \tau_{\alpha\beta}(x)v)]) = (x, \tau_{\alpha\beta}(x)v)$$
where $(x, \tau_{\alpha\beta}(x)v)$ is the unique element of $U_\alpha \times \mathbb{R}^k$ which is equivalent to $(x, v) \in U_\beta \times \mathbb{R}^k$ under $\sim$ (uniqueness follows from the previous sentence, since the equivalence class contains at most one element of $U_\alpha \times \mathbb{R}^k$).

The work above shows that $E$, together with the given data, satisfies the conditions of Lee's Lemma 10.6, so $E$ has a unique topology and smooth structure making it into a smooth rank-$k$ vector bundle whose local trivializations are the $\Phi_\alpha$; and of course the transition function between $\Phi_\alpha$ and $\Phi_\beta$ is $\tau_{\alpha\beta}$ by definition of transition function. $\square$

**Problem 2** (10-7)**.** Compute the transition function for $TS^2$ associated with the trivializations determined by stereographic coordinates.

*Solution.* Let $\sigma$ be the stereographic projection from the north pole and $\tilde{\sigma} = -\sigma(-x)$ the stereographic projection from the south pole, as per Lee's Problem 1-7. Then, we can compute

$$\begin{aligned}\tilde{\sigma}(\sigma^{-1}(u_1,u_2)) &= \tilde{\sigma}\left(\frac{u_1}{|u|^2+1}, \frac{u_2}{|u|^2+1}, \frac{|u|^2-1}{|u|^2+1}\right)\\ &= -\frac{1}{1+\frac{|u|^2-1}{|u|^2+1}}\left(-\frac{u_1}{|u|^2+1}, -\frac{u_2}{|u|^2+1}\right) = \frac{1}{|u|^2}(u_1,u_2)\end{aligned}$$

as the change of coordinates between $S^2$ minus the north pole $N$ and $S^2$ minus the south pole $S$, so that at each point $p$ of $S^2 - \{N\} - \{S\}$, the transition function is simply the map between tangent spaces induced by $\tilde{\sigma}\circ\sigma^{-1}$, namely the Jacobian

$$\frac{1}{(u_1^2+u_2^2)^2}\begin{pmatrix} u_2^2 - u_1^2 & -2u_1u_2 \\ -2u_1u_2 & u_1^2 - u_2^2\end{pmatrix}. \qquad \square$$

**Problem 3** (11-7)**.** In the following problems, $M$ and $N$ are smooth manifolds, $F\colon M \to N$ is a smooth map, and $\omega \in \mathfrak{X}^*(N)$. Compute $F^*\omega$ in each case.

(a) $M = N = \mathbb{R}^2$; $F(s,t) = (st, e^t)$; $\omega = x\mathrm{d}y - y\mathrm{d}x$.

(b) $M = \mathbb{R}^2$; $N = \mathbb{R}^3$; $F(\theta,\varphi) = ((\cos\varphi+2)\cos\theta, (\cos\varphi+2)\sin\theta, \sin\varphi)$; $\omega = z^2\mathrm{d}x$.

(c) $M = \{(s,t)\in\mathbb{R}^2 : s^2+t^2<1\}$; $N = \mathbb{R}^3 - \{0\}$; $F(s,t) = (s,t,\sqrt{1-s^2-t^2})$; $\omega = (1-x^2-y^2)\mathrm{d}z$.

*Solution.* For each $p \in M$ and $v \in T_pM$,

$$(F^*\omega)_p(v) = \omega_{F(p)}(\mathrm{d}F_p(v)).$$

(a) Since

$$\mathrm{d}F_{(s,t)} = \begin{pmatrix} t & s \\ 0 & e^t\end{pmatrix}$$

we can compute

$$(F^*\omega)_{(s,t)}\begin{pmatrix} a \\ b\end{pmatrix} = \omega_{F(s,t)}\begin{pmatrix} at+bs \\ be^t\end{pmatrix} = stbe^t - e^t(at+bs) = -te^t\cdot a + se^t(t-1)\cdot b,$$

so $F^*\omega = -te^t\mathrm{d}s + se^t(t-1)\mathrm{d}t$.

(b) Since

$$\mathrm{d}F_{(\theta,\varphi)} = \begin{pmatrix} -(\cos\varphi+2)\sin\theta & -\sin\varphi\cos\theta \\ (\cos\varphi+2)\cos\theta & -\sin\varphi\sin\theta \\ 0 & \cos\varphi\end{pmatrix}$$

we can compute

$$(F^*\omega)_{(\theta,\varphi)}\begin{pmatrix} a \\ b\end{pmatrix} = \omega_{F(\theta,\varphi)}\begin{pmatrix} -a(\cos\varphi+2)\sin\theta - b\sin\varphi\cos\theta \\ a(\cos\varphi+2)\cos\theta - b\sin\varphi\sin\theta \\ b\cos\varphi\end{pmatrix} = -\sin^2\varphi(\cos\varphi+2)\sin\theta\cdot a - \sin^3\varphi\cos\theta\cdot b$$

so $F^*\omega = -\sin^2\varphi(\cos\varphi+2)\sin\theta\mathrm{d}\theta - \sin^3\cos\theta\mathrm{d}\varphi$.

(c) Since

$$\mathrm{d}F_{(s,t)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{-s}{\sqrt{1-s^2-t^2}} & \frac{-t}{\sqrt{1-s^2-t^2}} \end{pmatrix}$$

we can compute

$$(F^*\omega)_{(s,t)} \begin{pmatrix} a \\ b \end{pmatrix} = \omega_{F(s,t)} \begin{pmatrix} a \\ b \\ \frac{-as-bt}{\sqrt{1-s^2-t^2}} \end{pmatrix} = -s\sqrt{1-s^2-t^2} \cdot a - t\sqrt{1-s^2-t^2} \cdot b$$

so $F^*\omega = -s\sqrt{1-s^2-t^2}\mathrm{d}s - t\sqrt{1-s^2-t^2}\mathrm{d}t$. □

**Problem 4** (11-9)**.** Let $f\colon \mathbb{R}^3 \to \mathbb{R}$ be the function $f(x,y,z) = x^2+y^2+z^2$, and let $F\colon \mathbb{R}^2 \to \mathbb{R}^3$ be

$$F(u,v) = \left(\frac{2u}{u^2+v^2+1}, \frac{2v}{u^2+v^2+1}, \frac{u^2+v^2-1}{u^2+v^2+1}\right),$$

the inverse of the stereographic projection. compute $F^*\mathrm{d}f$ and $\mathrm{d}(f \circ F)$ separately, and verify that they are equal.

*Solution.* Noting that $\mathrm{d}f = 2x\mathrm{d}x + 2y\mathrm{d}y + 2z\mathrm{d}z$, we can calculate

$$\begin{aligned} F^*(2x\mathrm{d}x) &= \frac{2u}{u^2+v^2+1}\left(\frac{2(-u^2+v^2+1)}{(u^2+v^2+1)^2}\mathrm{d}u - \frac{4uv}{(u^2+v^2+1)^2}\mathrm{d}v\right) \\ F^*(2y\mathrm{d}y) &= \frac{2v}{u^2+v^2+1}\left(-\frac{4uv}{(u^2+v^2+1)^2}\mathrm{d}u + \frac{2(u^2-v^2+1)}{(u^2+v^2+1)^2}\mathrm{d}v\right) \\ F^*(2z\mathrm{d}z) &= \frac{u^2+v^2-1}{u^2+v^2+1}\left(\frac{4u}{(u^2+v^2+1)^2}\mathrm{d}u + \frac{4u}{(u^2+v^2+1)^2}\mathrm{d}v\right). \end{aligned}$$

Summing, we find that the coefficient of $\mathrm{d}u$ is $(u^2+v^2+1)^{-3}$ times

$$4u(-u^2+v^2+1) - 8uv^2 + 4u(u^2+v^2-1) = 4u(2v^2) - 8uv^2 = 0,$$

and by symmetry the coefficient of $\mathrm{d}v$ is also zero, so $F^*\mathrm{d}f \equiv 0$.

On the other hand, $f \circ F = 1$ since $F$ sends points in $\mathbb{R}^2$ to the sphere in $\mathbb{R}^3$ by definition of (the inverse of) stereographic projection, implying $\mathrm{d}(f \circ F) = 0$, as expected from above. □